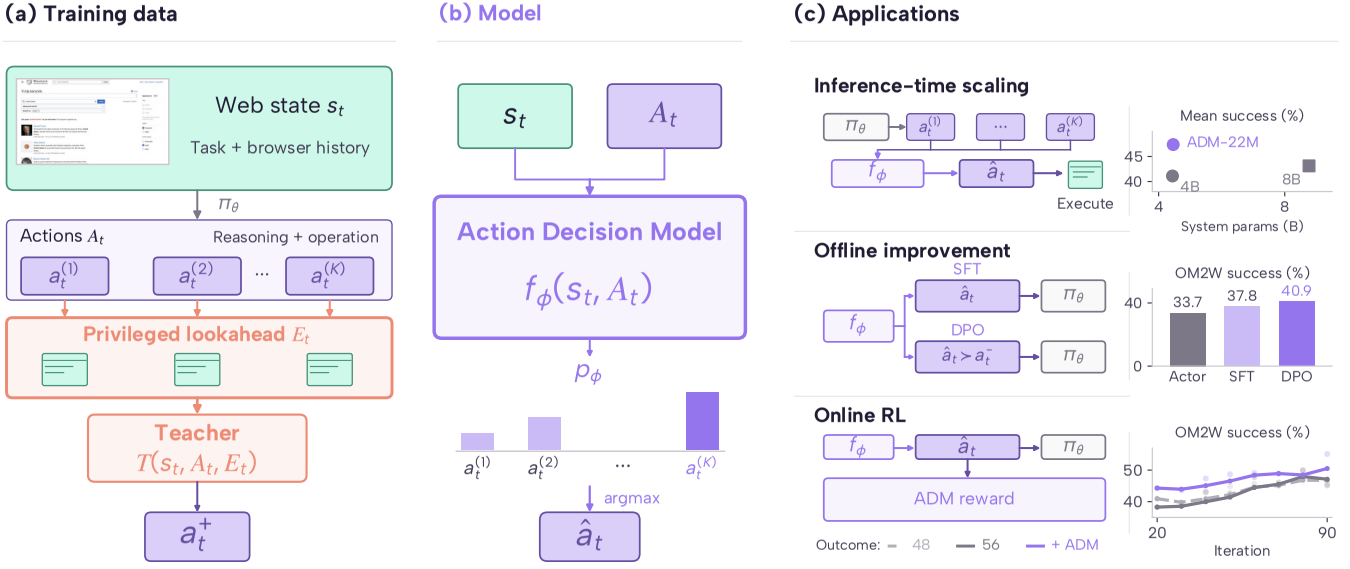
(a) Training data
Web state $s_t$
Task + browser history
$\pi_\theta$
Actions $A_t$
Reasoning + operation
$a_t^{(1)}$
$a_t^{(2)}$
...
$a_t^{(K)}$
Privileged lookahead $E_t$
Teacher
$T(s_t, A_t, E_t)$
$a_t^+$
(b) Model
$s_t$
$A_t$
Action Decision Model
$f_\phi(s_t, A_t)$
$p_\phi$
$a_t^{(1)}$
$a_t^{(2)}$
...
$a_t^{(K)}$
argmax
$\hat{a}_t$
(c) Applications
Inference-time scaling
$\pi_\theta$
$a_t^{(1)}$
...
$a_t^{(K)}$
$f_\phi$
$\hat{a}_t$
Execute
Mean success (%)
ADM-22M
45
40
4B
8B
4
8
System params (B)
Offline improvement
SFT
DPO
$f_\phi$
$\hat{a}_t$
$\hat{a}_t \succ a_t^-$
$\pi_\theta$
OM2W success (%)
40
0
33.7
37.8
40.9
Actor
SFT
DPO
Online RL
$f_\phi$
$\hat{a}_t$
$\pi_\theta$
ADM reward
OM2W success (%)
50
40
20
90
Iteration
Outcome:
48
56
+ ADM